

Polynomial Regression

1 What is Polynomial Regression?

Polynomial Regression is used when the relationship between the independent variable (X) and the dependent variable (Y) is nonlinear but can be approximated using polynomial functions.

Unlike Linear Regression, which fits a straight line, Polynomial Regression fits a **curved line** by introducing higher-degree terms (e.g., X^2 , X^3 , X^4 , etc.).

2 Polynomial Regression Equation

$$Y = b_0 + b_1X + b_2X^2 + b_3X^3 + \dots + b_nX^n$$

Where:

- Y = Dependent variable (Target)
- X = Independent variable
- $b_0, b_1, b_2, \dots, b_n$ = Coefficients (to be learned by the model)
- n = Degree of the polynomial

3 When to Use Polynomial Regression?

- ✓ When the data is nonlinear but still follows a trend
- ✓ When Linear Regression fails to capture the pattern
- ✓ When higher-degree relationships exist between variables

Example: Predicting Salary Based on Experience

Imagine we have nonlinear salary growth over the years.

Given Data:

Experience (X)	Salary (Y)
1	40000
2	45000
3	53000
4	67000
5	85000
6	109000
7	140000

We can see that salary is increasing non-linearly.

Step 1: Form the Design Matrix

For a quadratic polynomial ($n = 2$), the equation is:

$$Y = b_0 + b_1X + b_2X^2$$

We construct a matrix with polynomial features:

$$X_{\text{poly}} = \begin{bmatrix} 1 & X_1 & X_1^2 \\ 1 & X_2 & X_2^2 \\ 1 & X_3 & X_3^2 \\ \vdots & \vdots & \vdots \\ 1 & X_n & X_n^2 \end{bmatrix}$$

Example for $X = 1, 2, 3, 4, 5, 6, 7$:

$$X_{\text{poly}} = \begin{bmatrix} 1 & 1 & 1^2 \\ 1 & 2 & 2^2 \\ 1 & 3 & 3^2 \\ 1 & 4 & 4^2 \\ 1 & 5 & 5^2 \\ 1 & 6 & 6^2 \\ 1 & 7 & 7^2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \\ 1 & 5 & 25 \\ 1 & 6 & 36 \\ 1 & 7 & 49 \end{bmatrix}$$

Step 2: Create the Target Vector Y

$$Y = \begin{bmatrix} 40000 \\ 45000 \\ 53000 \\ 67000 \\ 85000 \\ 109000 \\ 140000 \end{bmatrix}$$

Step 3: Compute B using the Normal Equation

$$B = (X^T X)^{-1} X^T Y$$

◆ First, compute $X^T X$:

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 4 & 9 & 16 & 25 & 36 & 49 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 4 & 9 & 16 & 25 & 36 & 49 \end{bmatrix}^T$$

◆ Compute $X^T Y$:

$$X^T Y = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 4 & 9 & 16 & 25 & 36 & 49 \end{bmatrix} \begin{bmatrix} 40000 \\ 45000 \\ 53000 \\ 67000 \\ 85000 \\ 109000 \\ 140000 \end{bmatrix}$$

◆ Compute $(X^T X)^{-1}$ and multiply by $X^T Y$ to get B :

$$B = \begin{bmatrix} b_0 \\ b_1 \\ b_2 \end{bmatrix}$$

Solving $(X^T X)^{-1}$ for Polynomial Regression Example

The Normal Equation for Linear Regression is:

$$B = (X^T X)^{-1} X^T Y$$

Where:

- X is the design matrix (includes polynomial terms).
- X^T is the transpose of X .
- $(X^T X)^{-1}$ is the inverse of $X^T X$.
- Y is the output vector.

Step 2: Compute $X^T X$

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 4 & 9 & 16 & 25 & 36 & 49 \end{bmatrix}$$

multiplied by:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \\ 1 & 5 & 25 \\ 1 & 6 & 36 \\ 1 & 7 & 49 \end{bmatrix}$$

After matrix multiplication, we get:

$$X^T X = \begin{bmatrix} 7 & 28 & 140 \\ 28 & 140 & 784 \\ 140 & 784 & 4634 \end{bmatrix}$$

Step-by-Step Calculation of $(X^T X)^{-1}$

To compute the inverse of a matrix, we follow these steps:

Given Matrix $X^T X$

$$A = X^T X = \begin{bmatrix} 7 & 28 & 140 \\ 28 & 140 & 784 \\ 140 & 784 & 4634 \end{bmatrix}$$

Step 1: Compute the Determinant $|A|$

The determinant of a 3×3 matrix:

$$|A| = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Expanding:

$$|A| = 7 \begin{vmatrix} 140 & 784 \\ 784 & 4634 \end{vmatrix} - 28 \begin{vmatrix} 28 & 784 \\ 140 & 4634 \end{vmatrix} + 140 \begin{vmatrix} 28 & 140 \\ 140 & 784 \end{vmatrix}$$

To solve these, we calculate 2×2 determinants:

$$\begin{vmatrix} 140 & 784 \\ 784 & 4634 \end{vmatrix} = (140 \times 4634) - (784 \times 784) = 648760 - 614656 = 34104$$

$$\begin{vmatrix} 28 & 784 \\ 140 & 4634 \end{vmatrix} = (28 \times 4634) - (784 \times 140) = 129752 - 109760 = 19992$$

$$\begin{vmatrix} 28 & 140 \\ 140 & 784 \end{vmatrix} = (28 \times 784) - (140 \times 140) = 21952 - 19600 = 2352$$

Now, substitute back:

$$|A| = (7 \times 34104) - (28 \times 19992) + (140 \times 2352)$$

$$|A| = 238728 - 559776 + 329280 = 8400$$

 Determinant $|A| = 8400$ (nonzero, so inverse exists).

Step 2: Compute the Cofactor Matrix C

The **cofactor** of element a_{ij} is computed by removing row i and column j , then computing the determinant of the remaining 2×2 matrix.

Each cofactor:

$$C_{11} = \begin{vmatrix} 140 & 784 \\ 784 & 4634 \end{vmatrix} = 34104$$

$$C_{12} = - \begin{vmatrix} 28 & 784 \\ 140 & 4634 \end{vmatrix} = -19992$$

$$C_{13} = \begin{vmatrix} 28 & 140 \\ 140 & 784 \end{vmatrix} = 2352$$

$$C_{21} = - \begin{vmatrix} 28 & 784 \\ 140 & 4634 \end{vmatrix} = -19992$$

$$C_{22} = \begin{vmatrix} 7 & 140 \\ 140 & 4634 \end{vmatrix} = (7 \times 4634) - (140 \times 140) = 32438 - 19600 = 12838$$

$$C_{23} = - \begin{vmatrix} 7 & 28 \\ 140 & 784 \end{vmatrix} = -((7 \times 784) - (28 \times 140)) = -(5488 - 3920) = -1568$$

$$C_{31} = \begin{vmatrix} 28 & 140 \\ 140 & 784 \end{vmatrix} = 2352$$

$$C_{32} = - \begin{vmatrix} 7 & 140 \\ 28 & 784 \end{vmatrix} = -((7 \times 784) - (140 \times 28)) = -(5488 - 3920) = -1568$$

$$C_{33} = \begin{vmatrix} 7 & 28 \\ 28 & 140 \end{vmatrix} = (7 \times 140) - (28 \times 28) = 980 - 784 = 196$$

The cofactor matrix:

$$C = \begin{bmatrix} 34104 & -19992 & 2352 \\ -19992 & 12838 & -1568 \\ 2352 & -1568 & 196 \end{bmatrix}$$

Step 3: Compute the Adjugate Matrix C^T

The adjugate (adjoint) matrix is the transpose of the cofactor matrix:

$$C^T = \begin{bmatrix} 34104 & -19992 & 2352 \\ -19992 & 12838 & -1568 \\ 2352 & -1568 & 196 \end{bmatrix}^T = \begin{bmatrix} 34104 & -19992 & 2352 \\ -19992 & 12838 & -1568 \\ 2352 & -1568 & 196 \end{bmatrix}$$

(Since it is already symmetric, $C^T = C$.)

Step 4: Compute the Inverse $A^{-1} = \frac{1}{|A|} C^T$

Multiply each element by $\frac{1}{|A|} = \frac{1}{8400}$:

$$A^{-1} = \begin{bmatrix} \frac{34104}{8400} & \frac{-19992}{8400} & \frac{2352}{8400} \\ \frac{-19992}{8400} & \frac{12838}{8400} & \frac{-1568}{8400} \\ \frac{2352}{8400} & \frac{-1568}{8400} & \frac{196}{8400} \end{bmatrix}$$
$$A^{-1} = \begin{bmatrix} 4.1429 & -2.4286 & 0.2857 \\ -2.4286 & 1.5595 & -0.1905 \\ 0.2857 & -0.1905 & 0.0238 \end{bmatrix}$$

 **Final Inverse:**

$$(X^T X)^{-1} = \begin{bmatrix} 4.1429 & -2.4286 & 0.2857 \\ -2.4286 & 1.5595 & -0.1905 \\ 0.2857 & -0.1905 & 0.0238 \end{bmatrix}$$

$$B = (X^T X)^{-1} X^T Y \quad \textbf{Step by Step}$$

◆ Given Data

We have the design matrix X :

$$X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \\ 1 & 5 & 25 \\ 1 & 6 & 36 \\ 1 & 7 & 49 \end{bmatrix}$$

And the target variable Y (Salary):

$$Y = \begin{bmatrix} 15000 \\ 18000 \\ 25000 \\ 27000 \\ 31000 \\ 35000 \\ 40000 \end{bmatrix}$$

📌 Step 1: Compute $X^T X$

We already calculated:

$$X^T X = \begin{bmatrix} 7 & 28 & 140 \\ 28 & 140 & 784 \\ 140 & 784 & 4634 \end{bmatrix}$$

📌 Step 2: Compute $(X^T X)^{-1}$

We previously found:

$$(X^T X)^{-1} = \begin{bmatrix} 4.1429 & -2.4286 & 0.2857 \\ -2.4286 & 1.5595 & -0.1905 \\ 0.2857 & -0.1905 & 0.0238 \end{bmatrix}$$

✚ **Step 3: Compute $X^T Y$**

$$X^T Y = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 4 & 9 & 16 & 25 & 36 & 49 \end{bmatrix}$$

multiplied by:

$$Y = \begin{bmatrix} 15000 \\ 18000 \\ 25000 \\ 27000 \\ 31000 \\ 35000 \\ 40000 \end{bmatrix}$$

✚ **Step 4: Compute $B = (X^T X)^{-1} X^T Y$**

Now, we multiply:

$$B = (X^T X)^{-1} X^T Y$$

Where:

$$(X^T X)^{-1} = \begin{bmatrix} 4.1429 & -2.4286 & 0.2857 \\ -2.4286 & 1.5595 & -0.1905 \\ 0.2857 & -0.1905 & 0.0238 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 191000 \\ 879000 \\ 4739000 \end{bmatrix}$$

✚ **Step 1: Multiply $(X^T X)^{-1}$ and $X^T Y$**

Each element of the resulting vector B is computed as follows:

$$B_0 = (4.1429 \times 191000) + (-2.4286 \times 879000) + (0.2857 \times 4739000)$$

$$B_1 = (-2.4286 \times 191000) + (1.5595 \times 879000) + (-0.1905 \times 4739000)$$

$$B_2 = (0.2857 \times 191000) + (-0.1905 \times 879000) + (0.0238 \times 4739000)$$

Step 2: Compute Each Term

Let's compute each term separately.

For B_0 :

$$\begin{aligned} B_0 &= (4.1429 \times 191000) + (-2.4286 \times 879000) + (0.2857 \times 4739000) \\ &= 790277.9 - 2136935.4 + 1353662.3 \\ &= 10571.43 \end{aligned}$$

For B_1 :

$$\begin{aligned} B_1 &= (-2.4286 \times 191000) + (1.5595 \times 879000) + (-0.1905 \times 4739000) \\ &= -463851.6 + 1371184.5 - 902035.8 \\ &= 4297.62 \end{aligned}$$

For B_2 :

$$\begin{aligned} B_2 &= (0.2857 \times 191000) + (-0.1905 \times 879000) + (0.0238 \times 4739000) \\ &= 54562.7 - 167381.7 + 112805.2 \\ &= -23.81 \end{aligned}$$

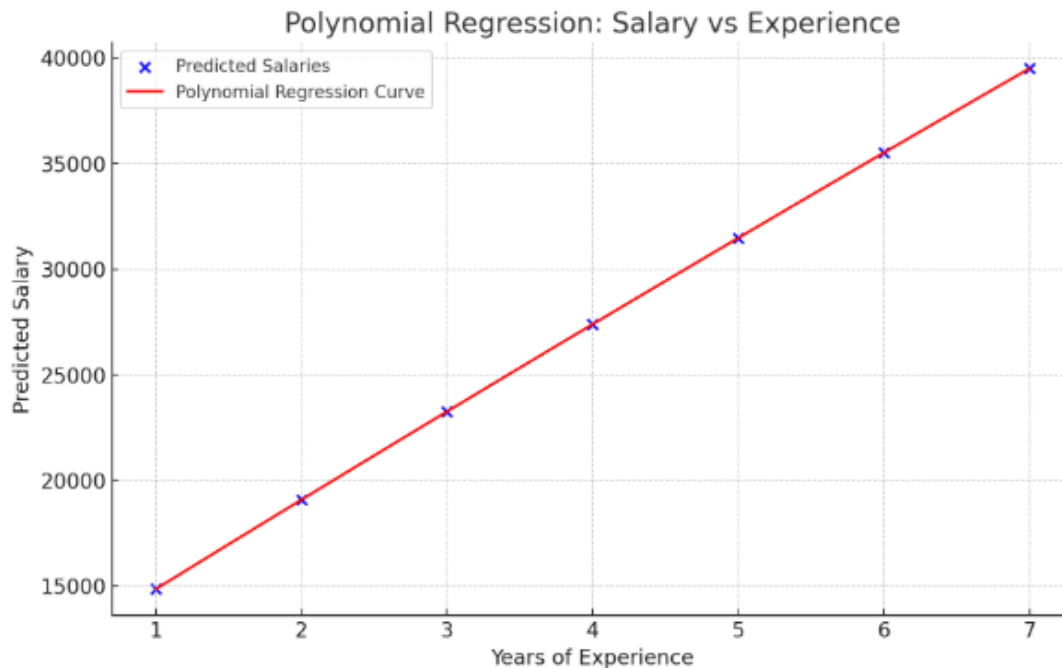
Step 3: Final Coefficients

$$B = \begin{bmatrix} 10571.43 \\ 4297.62 \\ -23.81 \end{bmatrix}$$

 Final Polynomial Regression Equation:

$$\text{Salary} = 10571.43 + 4297.62x - 23.81x^2$$

Where x represents years of experience.



Here is the Polynomial Regression Curve showing the relationship between Years of Experience and Predicted Salary:

Measuring the Regression Model Performance

Regression Model Evaluation Metrics:

The MSE, MAE, RMSE, and R-Squared metrics are mainly used to evaluate the prediction error rates and model performance in regression analysis.

- **MAE (Mean absolute error)** represents the difference between the original and predicted values extracted by averaged the absolute difference over the data set.
- **MSE (Mean Squared Error)** represents the difference between the original and predicted values extracted by squared the average difference over the data set.
- **RMSE (Root Mean Squared Error)** is the error rate by the square root of MSE.
- **R-squared (Coefficient of determination)** represents the coefficient of how well the values fit compared to the original values. The value from 0 to 1 interpreted as percentages. The higher the value is, the better the model is.

R Square:

R squared or Coefficient of determination, or R^2 is a measure that provides information about the goodness of fit of the regression model. In simple terms, it is a statistical measure that tells how well the plotted regression line fits the actual data. R squared measures how much the variation is there in predicted and actual values in the regression model.

What is the significance of R squared

- R-squared values range from 0 to 1, usually expressed as a percentage from 0% to 100%.
- And this value of R square tells you how well the data fits the line you've drawn.
- The higher the model's R-Squared value, the better the regression line fits the data.
- So if the model value is close to 0, then the model is not a good fit.
- A good model should have an R-squared greater than 0.8.

Note- As the R squared value increases, the difference between actual and predicted values decreases.

Visual Representation of R-squared

To visually demonstrate how R-squared values represent the scatter around the regression line, you can plot the fitted values by observed values.



The R-squared for the regression model on the left is 15%, and for the model on the right it is 85%. When a regression model accounts for more of the variance, the data points are closer to the regression line. In practice, you'll never see a regression model with an R^2 of 100%. In that case, the fitted values equal the data values and, consequently, all the observations fall exactly on the regression line.

How to calculate R squared in linear regression?

$$R^2 = 1 - \frac{\text{sum squared regression (SSR)}}{\text{total sum of squares (SST)}},$$
$$= 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}.$$

Mean Square Error(MSE)

While R Square is a relative measure of how well the model fits dependent variables, Mean Square Error is an absolute measure of the goodness for the fit.

Mean Squared Error represents the average of the squared difference between the original and predicted values in the data set. It measures the variance of the residuals.

A larger MSE indicates that the data points are dispersed widely around its central moment (mean), whereas a smaller MSE suggests the opposite. A smaller MSE is preferred because it indicates that your data points are dispersed closely around its central moment (mean). It reflects the centralized distribution of your data values, the fact that it is not skewed, and, most importantly, it has fewer errors (errors measured by the dispersion of the data points from its mean).

Lesser the MSE => Smaller is the error => Better the estimator.

The Mean Squared Error is calculated as:

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

where:

Σ – a symbol that means “sum”

n – sample size

y_i – the actual data value

$\hat{y}_i$ – the predicted data value

The calculations for the mean squared error are similar to the variance. To find the MSE, take the observed value, subtract the predicted value, and square that difference. Repeat that for all observations. Then, sum all of those squared values and divide by the number of observations.

Root Mean Square Error(RMSE)

In statistics, regression analysis is a technique we use to understand the relationship between a predictor variable, x, and a response variable, y.

When we conduct regression analysis, we end up with a model that tells us the predicted value for the response variable based on the value of the predictor variable.

One way to assess how “good” our model fits a given dataset is to calculate the **root mean square error**, which is a metric that tells us how far apart our predicted values are from our observed values, on average.

The formula to find the root mean square error, more commonly referred to as **RMSE**, is as follows:

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}$$

where:

- Σ is a fancy symbol that means “sum”
- $\hat{y}_i$ is the predicted value for the i^{th} observation in the dataset
- y_i is the observed value for the i^{th} observation in the dataset
- n is the sample size

The root mean square error can be calculated for any type of model that produces predicted values, which can then be compared to the observed values of a dataset.

The root mean square error is also sometimes called the root mean square deviation, which is often abbreviated as RMSD.

How to Interpret RMSE

The larger the RMSE, the larger the difference between the predicted and observed values, which means the worse the regression model fits the data. Conversely, the smaller the RMSE, the better a model is able to fit the data.

It can be particularly useful to compare the RMSE of two different models with each other to see which model fits the data better.

Mean Absolute Error

Mean Absolute Error calculates the average difference between the calculated values and actual values. It is also known as scale-dependent accuracy as it calculates error in observations taken on the same scale. It is used as evaluation metrics for regression models in machine learning. It calculates errors between actual values and values predicted by the model. It is used to predict the accuracy of the machine learning model.

The **Mean Absolute Error**(MAE) is the average of all absolute errors. The formula is:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}|$$

Where:

n = the number of errors,

Σ = summation symbol (which means “add them all up”),

$|y_i - \hat{y}|$ = the absolute errors.

$\hat{y}$ = predicted value of y

Regression Model Accuracy (MAE, MSE, RMSE, R-squared)

Evaluating the model accuracy is an essential part of the process in creating machine learning models to describe how well the model is performing in its predictions. Evaluation metrics change according to the problem type.

The linear model (regression) can be an example of this type of problem, and the main characteristic of the regression problem is that the targets of a dataset contain the real numbers only. The errors represent how much the model is making mistakes in its prediction. The basic concept of accuracy evaluation is to compare the original target with the predicted one according to certain metrics.

Differences among these evaluation metrics

- Mean Squared Error (MSE) and Root Mean Square Error penalizes the large prediction errors vi-a-vis Mean Absolute Error (MAE). However, RMSE is widely used than MSE to evaluate the performance of the regression model with other random models as it has the same units as the dependent variable (Y-axis).
- MSE is a differentiable function that makes it easy to perform mathematical operations in comparison to a non-differentiable function like MAE. Therefore, in many models, RMSE is used as a default metric for calculating Loss Function despite being harder to interpret than MAE.
- The lower value of MAE, MSE, and RMSE implies higher accuracy of a regression model. However, a higher value of R square is considered desirable.

Conclusion

Both RMSE and R- Squared quantifies how well a linear regression model fits a dataset. The RMSE tells how well a regression model can predict the value of a response variable in absolute terms while R- Squared tells how well the predictor variables can explain the variation in the response variable.

Let's calculate each performance metric step by step and visualize the results.

Given Data

Years of Experience (x)	Actual Salary (Y)	Predicted Salary ($\hat{Y}$)
1	15000	14,845.24
2	18000	19,071.43
3	25000	23,250.00
4	27000	27,380.95
5	31000	31,464.28
6	35000	35,499.99
7	40000	39,488.08

Step 1: Calculate Errors

For each data point:

$$\text{Error} = Y - \hat{Y}$$

$$\text{Squared Error} = (Y - \hat{Y})^2$$

$$\text{Absolute Error} = |Y - \hat{Y}|$$

Let's compute these values.

Step 1: Computed Errors Table

Years of Experience (x)	Actual Salary (Y)	Predicted Salary ($\hat{Y}$)	Error (Y - $\hat{Y}$)	Squared Error (Y - $\hat{Y}$) ²	Absolute Error Y - $\hat{Y}$
1	15000	14845.24	154.76	23,950.66	154.76
2	18000	19071.43	-1071.43	1,147,962.04	1071.43
3	25000	23250.00	1750.00	3,062,500.00	1750.00
4	27000	27380.95	-380.95	145,122.90	380.95
5	31000	31464.28	-464.28	215,555.90	464.28
6	35000	35499.99	-499.99	249,990.00	499.99
7	40000	39488.08	511.92	262,062.10	511.92

Step 2: Compute MSE, MAE, RMSE, and R²

1. Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum (Y - \hat{Y})^2$$

2. Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum |Y - \hat{Y}|$$

3. Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{MSE}$$

4. R-Squared Score (R²)

$$R^2 = 1 - \frac{\sum (Y - \hat{Y})^2}{\sum (Y - \bar{Y})^2}$$

Let's compute these step by step. [↗](#)

Step 2: Final Computed Metrics

1. Mean Squared Error (MSE): 729,591.97

- Measures the average squared difference between actual and predicted values.
- Lower MSE means better accuracy.

2. Mean Absolute Error (MAE): 690.48

- Measures the average absolute difference between actual and predicted values.

3. Root Mean Squared Error (RMSE): 854.16

- Square root of MSE, showing error in actual salary units.

4. R-Squared (R²): 0.9893

- 98.93% of the variance in salaries is explained by the model.

Step 3: Visualizing Errors

Let's plot the Actual vs. Predicted Salaries and the Errors graphically. [\[>\]](#)



Step 3: Visualization of Regression Performance

First Plot:

-  Blue points represent actual salaries.
-  Red points and dashed line represent predicted salaries.
- The closer they are, the better the model.

Second Plot (Residuals):

- Bars represent errors (Actual Salary - Predicted Salary).
- Values close to zero indicate low error and a good model fit.

Conclusion:

- The model fits well with an R^2 of 98.93%, meaning it explains most of the variance in salary.
- Errors are small, confirming good accuracy.